

Irish Sign Language in a Virtual Reality Environment

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Figure 1: Irish Sign Language in VR

ABSTRACT

This paper describes a study whereby groups of users are tasked with learning Irish Sign Language (ISL) gestures in distinct environments and performing a comparison of outcomes to determine the general effectiveness of each method. One groups of users are tasked with learning via traditional tuition methods, and the second groups is placed in a virtual reality environment with a custom training module for teaching ISL gestures. This paper discusses such a study and details the processes, motivations and potential future work.

Index Terms: Human-centered computing—Accessibility—Accessibility systems and tools; Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction—Interaction paradigms—Collaborative interaction; Applied computing—Arts and humanities—Language translation

1 INTRODUCTION

Recent advancements in virtual reality have allowed the technology to expand into a huge variety of sectors and a multitude of use cases, from manufacturing and healthcare, to education, to name a few. Modern virtual reality headsets such as the Oculus Quest are equipped with real-time hand tracking, facilitating natural interactions and expanding the possibilities for useful applications.

In this paper, we perform a comparative study around the concept of using virtual reality and hand tracking technologies for teaching Irish Sign Language, a language which is used by over 40,000 people [1]. The study aims to be an initial step towards the concept, and acquire a general indicator as to the potential effectiveness of virtual reality as an educational tool in the area of Irish Sign Language. For this study, a custom virtual reality application has

been built for the Oculus Quest using Unreal Engine 4, making use of the device's built-in hand tracking capabilities along with purpose-built Irish Sign Language recognition functions to achieve the intended outcomes.

2 MOTIVATION

More than five percent of the global population suffer from some form of hearing disability, meaning that over 70 million people worldwide are users of sign language as a primary means of communication [2]. There are more than 200 unique sign languages in use across the globe, with their own distinct grammar.

Much like spoken languages, sign language tuition has evolved over time, with e-Learning courses in particular becoming more prevalent. In recent years, the area of spoken language learning has tended towards online, mobile solutions such as Duolingo and other similar applications.

However, sign language is distinct from spoken language in that it is not only a language of the hands, but also of the face and of the body. This means that the learning requirements of sign language are entirely different, and thus unique teaching solutions become necessary.

Sign languages are often learned in classes, whether group or individual sessions, but due to the global pandemic and its guidelines on classroom environments, it presents its own challenges with such arrangements. This has been a contributor to the rise of e-Learning among sign language learning. Video conferencing tools such as Zoom are increasingly used to offer classes for those seeking to learn sign languages, and, while a useful tool, can struggle to offer the presence and spatial awareness to effectively communicate via sign language due to their two-dimensional presentation. This consideration in particular drove the motivation for this study.

3 STAKEHOLDERS

This project primarily aims to assist those seeking to begin learning Irish Sign Language, whether as a primary or supplementary form of communication. It will be used to help those who are hard of hearing

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and/or their family, friends and caregivers to begin their journey towards proficiency with Irish Sign Language. The application will help with facilitate immersive learning in virtual reality to make use of hand tracking in three-dimensional space and ease the difficulty of learning such an expressive language when compared to traditional online classes which have become more prevalent in recent times.

4 METHODOLOGY

The methodology of this study involves performing a comparison in the effectiveness of teaching methods for sign language. While the ideal method for learning sign language would be a one-to-one interaction with another person in the same room, this study argues that virtual reality can be an effective substitute, and can offer a higher quality learning experience in comparison with methods confined to two dimensions, such as online learning via Zoom, or traditional teaching methods such as textbooks. With an emphasis on data retention, enhanced user attentiveness and interaction, as well as improved immersion of the end user.

4.1 Baseline Testing

To determine a baseline for comparison with the virtual reality application, a simple test is used for the first user group. This group is tasked with learning alphabetical gestures for Irish Sign Language using what can be deemed a more traditional teaching method.

This involves users being given an image displaying the hand poses of each letter of the ISL alphabet on paper, and being given a short period of time to learn the hand pose. Once the given time has elapsed, users are tested by being asked to perform the correct gesture for a series of letters. The outcome of this test is stored and averaged to determine a base for comparison with users of the virtual reality application, as described in the following section.

4.2 Virtual Reality Experiment

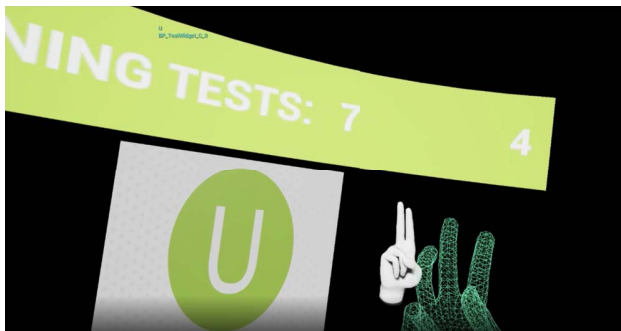


Figure 2: Virtual reality testing procedure

Users are placed within a custom virtual reality training application for the Oculus Quest. The purpose of the application is to provide a basic learning and testing environment for use within the study to help ascertain the effectiveness of virtual reality integration within the sign language learning process.

Built-in hand tracking is used to track the user's gestures and analysed to recognise predefined sign language gesticulations representing alphabetical characters of Irish Sign Language. Each hand pose representing alphabetical characters was prerecorded for use within the software's active gesture recognition system.

Users within the study are given a period of time to learn and practice the poses for each letter of the alphabet within the virtual environment, with an automated three-dimensional demonstration of gestures within the environment to assist with the process.

Once preparation time has elapsed, users are given a simple test in which a series of letters are chosen at random, and the user must

perform the gesture corresponding to that letter within the allotted time. User score is tracked throughout, along with any incorrect responses in order to determine potential tracking issues.

The user scores are then aggregated at the conclusion of the study and used to gauge effectiveness between tuition methods. The overall goal is to achieve at least a similar result level between groups as a starting point, with future work, outlined in the next section, intended to push beyond the limitations of traditional teaching methods such as online screen-based tuition and textbooks which are often used within the context of learning Irish Sign Language.

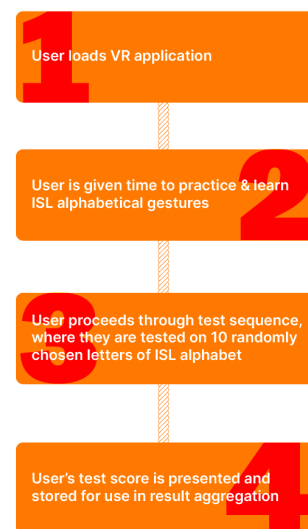


Figure 3: Virtual reality application user flow

5 THE FUTURE

This study represents merely an introductory step into the area of sign languages in virtual reality. The introduction of hand tracking into widely available consumer devices dramatically increased the potential for such applications, and it is reasonable to expect that further technological advances, including tracking accuracy, could offer a vast improvement over the current state of the art and methodologies. Additional work would also look at measuring the accuracy of software-based gesture recognition and system improvements that can be made in this area.

This study focused purely on static gestures representing alphabetical characters of Irish Sign Language, and future work would seek to incorporate motion-based multi-gesture recognition as an addition to current work. This would mean the addition of words and phrases to the learning environment, and that a more advanced comparative study can be performed to determine the effectiveness of virtual reality as a teaching tool for sign language.

ACKNOWLEDGMENTS

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